ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 2 Mechatronics paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, examples and neat sketches.

SECTION A

QUESTION 1

1.1 1.1.1 B ✓ 1.1.2 D ✓ 1.1.3 B ✓ 1.1.4 C ✓ 1.1.5 A ✓ 1.1.6 B ✓ 1.1.7 D ✓ 1.1.8 A ✓ 1.1.9 C ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 True ✓ 1.2.8 False ✓ 1.2.9 True ✓ 1.2.10 True ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 J ✓ 1.3.3 G ✓ 1.3.4 F ✓ 1.3.5 H ✓ 1.3.6 C ✓ 1.3.7 D ✓ 1.3.8 E ✓ 1.3.9 B ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Front view – shows the height and width / main shape seen from the front ✓✓; top (plan) view – shows the width and depth seen from above ✓✓; side (end) view – shows the height and depth seen from the side ✓✓ (6)

2.2 Any FOUR drawn and named ✓✓✓✓: thick continuous (visible outline); thin continuous (dimension / projection lines); short dashes (hidden detail); chain line – long and short dashes (centre line); thin wavy / zig-zag (break line); thick chain (cutting plane) (4)

2.3 $\text{Reading} = \text{main scale} + (\text{vernier division} \times \text{least count})$ ✓ $= 23 + (14 \times 0{,}02)$ ✓ $= 23{,}28\ \text{mm}$ ✓ (3)

2.4 Any THREE ✓✓✓: keep clean and dry; store in its case; do not drop or force; do not use as a hammer or clamp; check zero before use; do not measure moving or hot parts (3)

[16]

QUESTION 3

3.1 (a) aluminium ✓ – light and strong, corrosion resistant ✓ (b) nylon / acetal (plastic) ✓ – quiet, self-lubricating ✓ (c) ABS / polycarbonate plastic or sheet steel ✓ – insulating / tough / protects against dust ✓ (accept other motivated answers) (6)

3.2 Permanent joints cannot be taken apart without damaging the parts ✓ – e.g. welding, riveting, soldering, gluing (any TWO) ✓✓; temporary joints can be taken apart and reused ✓ – e.g. nuts and bolts, screws, pins, clips, keys (any TWO) ✓✓ (6)

3.3 Vibration can make nuts and bolts work loose ✓; a locknut / thread locker keeps the fastener tight so parts do not fall off or become misaligned ✓. (2)

[14]

QUESTION 4

4.1.1 $W = m \times g$ ✓ $= 60 \times 9{,}81 = 588{,}6\ N$ ✓ (2)

4.1.2 $\text{Effort arm} = 1{,}2 - 0{,}3 = 0{,}9\ m$ ✓; $E \times 0{,}9 = 588{,}6 \times 0{,}3$ ✓ $\rightarrow E = 196{,}2\ N$ ✓ (3)

4.1.3 $MA = \text{load} \div \text{effort}$ ✓ $= 588{,}6 \div 196{,}2 = 3$ ✓ (2)

4.2.1 $N_2 = N_1 \times T_1 \div T_2$ ✓ $= 1\,500 \times 20 \div 75$ ✓ $= 400\ r/min$ ✓ (3)

4.2.2 Larger ✓ – the gears reduce the speed, so the torque increases by the same ratio (power stays about the same) ✓. (2)

4.3 $W = F \times d = m \times g \times d$ ✓ $= 5 \times 9{,}81 \times 0{,}8$ ✓ $= 39{,}24\ J$ ✓; $P = W \div t$ ✓ $= 39{,}24 \div 2 = 19{,}62\ W$ ✓ (5)

4.4 Any TWO (mechanism ✓ + example ✓): rack and pinion – moving a gripper or a camera slide; lead screw / ball screw and nut – 3D printer or CNC axis; crank and slider – pump or pick-and-place pusher; cam and follower – stamping / valve movement (4)

[21]

QUESTION 5

5.1 Open loop: the controller sends a command without checking the output ✓✓ – e.g. a timed conveyor or stepper motor moved a set number of steps without checking ✓; closed loop: the output is measured and fed back so the controller can correct errors ✓✓ – e.g. a servo motor, line-following robot, speed control with an encoder ✓ (6)

5.2 Correct loop drawn ✓; input / set point and summing point ✓; controller ✓; actuator (motor driver / motor) ✓; process / output (motor speed) ✓; sensor (encoder) with feedback line back to the summing point ✓ (6)

5.3 It detects when the moving part reaches the end of its travel ✓ and signals the controller to stop / reverse, preventing damage and over-travel ✓. (2)

5.4 Any FIVE ✓✓✓✓✓: switch off and isolate the supply; lock out and tag out; test that the circuit is dead before touching; use insulated tools; do not work with wet hands; wear suitable PPE; only qualified / authorised people work on live panels; keep the panel closed when not working on it; never bypass safety devices (5)

[19]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE